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Maths worked solutions

Q16 · Length of the two tangents

Let T be the midpoint of MN. Since OM = ON and PM = PN, both O and P lie on the perpendicular bisector of MN. Thus O, T and P are collinear and MT = 3.

In right triangle OTM, OT = $\sqrt{(5^2 - 3^2)} = 4$. Also OM $\perp$ PM, since PM is a tangent. The acute angles TOM and TMP are equal because their corresponding arms are perpendicular. Therefore $\triangle$OTM ~ $\triangle$MTP by AA.

PM/OM = MT/OT = 3/4, so PM = 15/4. The two tangents from P have equal length.

PM + PN = 15/2 = 7.5 units.

Q17 · Perimeter of the equilateral triangle

Let T be the point where the smaller circle touches QR. Then OT $\perp$ QR. Since QR is a chord of the larger circle, T bisects QR.

Let R_0 and r be the outer and inner radii. The area difference gives $R_0^2 - r^2 = 616 \div (22/7) = 196$.

In right triangle OTQ, $QT^2 = OQ^2 - OT^2 = R_0^2 - r^2 = 196$. Thus QT = 14 cm and QR = 28 cm.

Perimeter = 3 × 28 = 84 cm.

Q12 · Percentage of women

Let p be the fraction of women, so 1 − p is the fraction of men. The combined mean gives 38(1 − p) + 30p = 35. Thus 8p = 3 and p = 3/8.

Percentage of women = 37.5%, option (D).

Q23 · Trigonometric expression

Since $\tan\theta = 1/\sqrt{5}$, $\cot\theta = \sqrt{5}$. Thus $\sec^2\theta = 1 + \tan^2\theta = 6/5$ and $\text{cosec}^2\theta = 1 + \cot^2\theta = 6$.

$(\text{cosec}^2\theta - \sec^2\theta)/(\text{cosec}^2\theta + \sec^2\theta) = (6 - 6/5)/(6 + 6/5) = 2/3$.

Maths worked solutions

Example 41 · Prove EL = 2BL

Since BC ∥ DE, B, M and E are collinear, and C, M and D are collinear, △BMC ~ △EMD by AA. As CM = DM, the ratio BC/DE = CM/DM = 1. Thus DE = BC.

Also AD = BC, so AE = AD + DE = 2BC. Since AE ∥ BC, △LBC ~ △LEA by AA.

Therefore EL/BL = AE/BC = 2, so EL = 2BL.

Q21 · Real and equal roots

The discriminant is zero. Dividing it by 4 gives $(a^2 - bc)^2 - (c^2 - ab)(b^2 - ac) = 0$.

Expanding, the left side is $a^4 - 2a^2bc + b^2c^2 - (b^2c^2 - ac^3 - ab^3 + a^2bc) = a^4 + ab^3 + ac^3 - 3a^2bc = a(a^3 + b^3 + c^3 - 3abc)$.

Hence $a = 0$ or $a^3 + b^3 + c^3 = 3abc$.

Q22 · The second quadratic has no real roots

The discriminant of $x^2 + 2cx + ab = 0$ is $4(c^2 - ab)$. Real, unequal roots imply that this is positive, so $c^2 > ab$.

The second discriminant is $4(a + b)^2 - 4(a^2 + b^2 + 2c^2) = 8(ab - c^2)$, which is negative.

Therefore the second equation has no real roots.

Q31(B) · Trigonometric proof

The given equation says $1/(\sin x - \cos x) = 1/(\sqrt{2} \sin x)$. Equating denominators gives $\cos x = (1 - \sqrt{2}) \sin x$.

Consequently $\sin x + \cos x = (2 - \sqrt{2}) \sin x = -\sqrt{2} \cos x$.

Taking reciprocals and squaring gives $[1/(\sin x + \cos x)]^2 = \sec^2 x/2$.

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Maths worked solutions

Q22 · Prove $DP^2 = BD \times CD$

Because PQ ∥ BR, △DPQ ~ △DBR by AA. Thus DP/DB = DQ/DR. Because CQ ∥ PR, △DCQ ~ △DPR by AA. Thus DC/DP = DQ/DR.

Equating ratios, DP/DB = DC/DP, so $DP^2 = BD \times CD$.

For PD = 12 cm, BD × CD = 144 cm^2.

Q23 · Prove $9AD^2 = 7AB^2$

Let a be the side length and let M be the foot of the altitude from A to BC. Since ABC is equilateral, BM = a/2. Also BD = a/3, so DM = a/6.

In right triangle ABM, $AM^2 = a^2 - (a/2)^2 = 3a^2/4$. In right triangle ADM, $AD^2 = AM^2 + DM^2 = 3a^2/4 + a^2/36 = 7a^2/9$.

Since a = AB, we get $9AD^2 = 7AB^2$.

Q34 · Prove OC ∥ SR

Since PS ∥ AB, △OPS ~ △OAB, so OS/OB = PS/AB. Likewise QR ∥ AB, so △CQR ~ △CAB and CR/CB = QR/AB.

Opposite sides of parallelogram PQRS have equal lengths, so PS = QR. Therefore OS/OB = CR/CB. Subtracting each ratio from 1 gives BS/BO = BR/BC.

By the converse of the basic proportionality theorem in △BOC, SR ∥ OC.

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Maths worked solutions

Q34 · Prove BD/CD = BF/CE

Draw CG ∥ DF meeting BA at G. Since E is the midpoint of AC and EF ∥ CG, the midpoint theorem in △ACG gives AF = FG.

Since ∠AEF = ∠AFE, their opposite sides are equal, so AF = AE. Also AE = CE. Hence FG = CE.

Since CG ∥ DF, △BCG ~ △BDF. Thus BC/BD = BG/BF. Subtracting both ratios from 1 gives CD/BD = FG/BF = CE/BF.

Taking reciprocals gives BD/CD = BF/CE.

(ii) · Ratio of the seventh and tenth terms of an A.P.

The printed condition $S_m:S_n = (2m + 1):(3n + 1)$ does not specify m and n. The following two choices give different answers.

If m = 1 and n = 2, the A.P. 3, 4, 5, … gives $S_1:S_2 = 3:7$ and $t_7:t_{10} = 9:12 = 3:4$.

If m = 1 and n = 3, the A.P. with first term 3 and common difference 1/3 gives $S_1:S_3 = 3:10$ and $t_7:t_{10} = 5:6$.

Interpreting the condition as applying to every pair also fails. For m = n = 1, it would require $S_1:S_1 = 3:4$.

The printed question does not determine a unique ratio.